

KANAK TANWAR

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Experience

Backend Intern | Vetic, Gurgaon, India

April 2025 – Present

- Built a **config-driven DSL rule engine** for appointment exemptions, using declarative rule files supporting multiple operators to enforce outcomes, reducing revenue leakage to **0%** while enabling non-code rule updates.
- Architected an automated **clinic day-closing and revenue reconciliation system**, replacing manual workflows and reducing processing time from **2 hours to 5 minutes**, while enabling systematic detection of revenue leakages.
- Scaled **payment processing pipelines** by integrating POS systems and Razorpay payment links, handling **80K+ monthly transactions**, improving payment adherence from **90% to 99.7%** and reducing billing mismatches by **40%**.
- Designed a **centralized cash management service** for multi-clinic operations, ensuring consistent tracking of daily cash flows and reducing reconciliation discrepancies across locations.
- Worked with **asynchronous processing pipeline** using **Cron jobs and RabbitMQ**, decoupling critical workflows and improving system reliability and throughput under peak loads.
- Developed an **LLM-assisted OCR pipeline** for extracting structured data from pet vaccination records, reducing manual data entry effort by **85%** and improving data accuracy.
- Improved system correctness under concurrency by implementing **Redis-based rate limiting, distributed locks, and MongoDB transactions**.
- Reduced API latency by **50%** by eliminating **N+1 queries**, optimizing SQL execution, and introducing **caching (local + Redis)**.

AI Intern | Mentorpal.ai

June 2024 – July 2024

- Developed a Chrome extension recommending personalized LinkedIn messages to prospective leads.
- Built internal resume QA tool using LangChain and LLMs, boosting QA efficiency by 40%.

Projects

Pair Programmer (pp)

[GitHub](#)

- Built an **agentic CLI coding assistant** with a rich TUI, enabling interactive codebase exploration, reasoning, and modification using LLMs.
- Designed a **tool-augmented agent architecture** with file system primitives (`read_file`, `apply_patch`, `grep`) to support structured multi-step edits, navigation, and autonomous workflows.
- Implemented **context persistence (memory + task tracking)** and a **provider-agnostic LLM interface** with streaming responses and dynamic model selection for flexible, long-running sessions.

Git Chat

[GitHub](#)

- Built a **RAG-based system** enabling conversational Q/A over public GitHub repositories using LLMs.
- Designed a **multi-model backend architecture** with FastAPI, integrating **Weaviate (vector search)**, **MongoDB (metadata)**, and **LangChain (retrieval orchestration)**.

Skills

- Languages & Frameworks** - Python, Go, FastAPI, Django, Flask, Gin
- Databases & ORM** - PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Beanie, Alembic, Golang-migrate
- DevOps & Tools** - Git, Linux, Docker, AWS, Grafana, Loki, Prometheus
- Misc** - REST APIs, RabbitMQ, Cron Jobs

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Graduation
B.Tech - CSE	Dronacharya College Of Engineering	86%	Ongoing
Senior Secondary	Manav Rachna International School	96.0%	May 2022
Secondary	Manav Rachna International School	97.6%	May 2020

Achievements

- ICPC Asia West Amritapuri 2023** Regionalist
- LeetCode max rating **1968** (Knight), CodeChef max rating **1684** (3★)
- Global Rank **159** in CodeChef Starters 155
- Qualified GATE 2025 — DA (AIR 702), CS (AIR 3050)